F16-SR-001
11kV FLOTATION SWITCHROOM

F16-SW-011
FLOTATION 11kV SWITCHBOARD

2000-F16-SCM-EL-10101

NOTE 2

A01

TO INCOMER PROTECTION RELAY
11kV 1250A 3-E 25kA FOR 3s
110V AC SUPPLY

CONT'D ON SHEET 2 DRG 2000-F16-SLD-EL-10002

2A
630A

NOTE 2

VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

A02	A03	A04	A05	A06
NOTE 3	NOTE 3	NOTE 3	NOTE 3	NOTE 3
100E	100E	100E	100E	100E
CLOSE	CLOSE	CLOSE	CLOSE	CLOSE
50 50G 81 AFD	50 50G 81 AFD	50 50G 81 AFD	50 50G 81 AFD	50 50G 81 AFD
51 51G	51 51G	51 51G	51 51G	51 51G
PR SEL 751	PR SEL 751	PR SEL 751	PR SEL 751	PR SEL 751
50/1A 2.5VA 5P20	50/1A 2.5VA 5P20	50/1A 2.5VA 5P20	50/1A 2.5VA 5P20	50/1A 2.5VA 5P20
TEST	TEST	TEST	TEST	TEST
50/1A 1.5VA 10P10	50/1A 1.5VA 10P10	50/1A 1.5VA 10P10	50/1A 1.5VA 10P10	50/1A 1.5VA 10P10
TEST	TEST	TEST	TEST	TEST
VSD CONTACTOR CLOSE PERMISSIVE	VSD CONTACTOR CLOSE PERMISSIVE	VSD CONTACTOR CLOSE PERMISSIVE	VSD CONTACTOR CLOSE PERMISSIVE	VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY	VSD HEALTHY	VSD HEALTHY	VSD HEALTHY	VSD HEALTHY
VSD FAULT	VSD FAULT	VSD FAULT	VSD FAULT	VSD FAULT
F16AG421-VS ROUGHER FLOTATION CELL 1 VSD 11/3.3kV 500kW	F16AG422-VS ROUGHER FLOTATION CELL 2 VSD 11/3.3kV 500kW	F16AG423-VS ROUGHER FLOTATION CELL 3 VSD 11/3.3kV 500kW	F16AG424-VS SCAVENGER FLOTATION CELL 1 VSD 11/3.3kV 500kW	F16AG425-VS SCAVENGER FLOTATION CELL 2 VSD 11/3.3kV 500kW
M 3~ RTD 500 kW F16-AG-421 ROUGHER FLOTATION CELL 1 AGITATOR 2000-F16-SCM-EL-10103	M 3~ RTD 500 kW F16-AG-422 ROUGHER FLOTATION CELL 2 AGITATOR 2000-F16-SCM-EL-10105	M 3~ RTD 500 kW F16-AG-423 ROUGHER FLOTATION CELL 3 AGITATOR 2000-F16-SCM-EL-10107	M 3~ RTD 500 kW F16-AG-424 SCAVENGER FLOTATION CELL 1 AGITATOR 2000-F16-SCM-EL-10109	M 3~ RTD 500 kW F16-AG-425 SCAVENGER FLOTATION CELL 2 AGITATOR 2000-F16-SCM-EL-10111

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

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Single line diagram sheet 1